GALOIS ACTION ON IRREDUCIBLE COMPONENTS OF A LOCALLY FINITE TYPE SCHEME: CONTINUITY, FINITENESS OF THE DEFINING EXTENSION, AND ORBITS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is complete as written and ends with the identification W = T; no step is referred to the reader.

Editorial note. The counted spans carry 4 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-galois-action-irreducible-components-locally-finite-type is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Locally of finite type means affine opens are finite type algebras. Morphisms, Lemma morphisms-lemma-locally-finite-type-characterize, invoked at proof lines 1815-1816. Supplies that the affine chart A is a finite type k-algebra, hence that A tensor the closure is Noetherian.
- Irreducible components of the base change and their defining ideals. Same chapter, Lemma lemma-galois-action-irreducible-components, invoked at proof lines 1827-1831. Supplies that a component invariant under an open subgroup comes from a component over a finite subextension.
- Descent of Galois invariant closed subsets. Same chapter, Lemma lemma-closed-fixed-by-Galois, invoked at proof lines 1842-1843. Supplies that a Galois invariant closed subset of the base change is itself a base change of a closed subset of X.
- Existence of a component of the base change mapping onto a given component. Same chapter, Lemma lemma-image-irreducible, invoked at proof lines 1835-1837. Supplies the component used in part (3).

Source reference. Original lemma 038B: sources/stacks-varieties/pinned-original-source.tex, source lines 899–899. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 038J: sources/stacks-varieties/pinned-original-source.tex, source lines 1700–1700. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 04KX: sources/stacks-varieties/pinned-original-source.tex, source lines 1622–1622. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01T2: sources/stacks-chow/tags.txt, source lines 2359–2359. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let k be a field. Assume $X \to \mathrm{Spec}(k)$ locally of finite type. In this case*

(1) *the action*
$$Gal(\overline{k}/k)^{opp} \times IrredComp(X_{\overline{k}}) \longrightarrow IrredComp(X_{\overline{k}})$$
is continuous if we give $IrredComp(X_{\overline{k}})$ the discrete topology,
(2) *every irreducible component of $X_{\overline{k}}$ can be defined over a finite extension of k, and*
(3) *given any irreducible component $T \subset X$ the scheme $T_{\overline{k}}$ is a finite union of irreducible components of $X_{\overline{k}}$ which are all in the same $Gal(\overline{k}/k)$-orbit.*

Proof. Let $\overline{T}$ be an irreducible component of $X_{\overline{k}}$. We may choose an affine open $U \subset X$ such that $\overline{T} \cap U_{\overline{k}}$ is not empty. Write $U = \mathrm{Spec}(A)$, so A is a finite type k-algebra, see Morphisms, Lemma 01T2. Hence $A_{\overline{k}}$ is a finite type $\overline{k}$-algebra, and in particular Noetherian. Let $\mathfrak{p} = (f_1, \ldots, f_n)$ be the prime ideal corresponding to $\overline{T} \cap U_{\overline{k}}$. Since $A_{\overline{k}} = A \otimes_k \overline{k}$ we see that there exists a finite subextension $\overline{k}/k'/k$ such that each $f_i \in A_{k'}$. It is clear that $\mathrm{Gal}(\overline{k}/k')$ fixes $\overline{T}$, which proves (1).

Part (2) follows by applying Lemma 038J (1) to the situation over k' which implies the irreducible component $\overline{T}$ is of the form $T'_{\overline{k}}$ for some irreducible $T' \subset X_{k'}$.

To prove (3), let $T \subset X$ be an irreducible component. Choose an irreducible component $\overline{T} \subset X_{\overline{k}}$ which maps to T, see Lemma 04KX. By the above the orbit of $\overline{T}$ is finite, say it is $\overline{T}_1, \ldots, \overline{T}_n$. Then $\overline{T}_1 \cup \ldots \cup \overline{T}_n$ is a $\mathrm{Gal}(\overline{k}/k)$-invariant closed subset of $X_{\overline{k}}$ hence of the form $W_{\overline{k}}$ for some $W \subset X$ closed by Lemma 038B. Clearly $W = T$ and we win. □